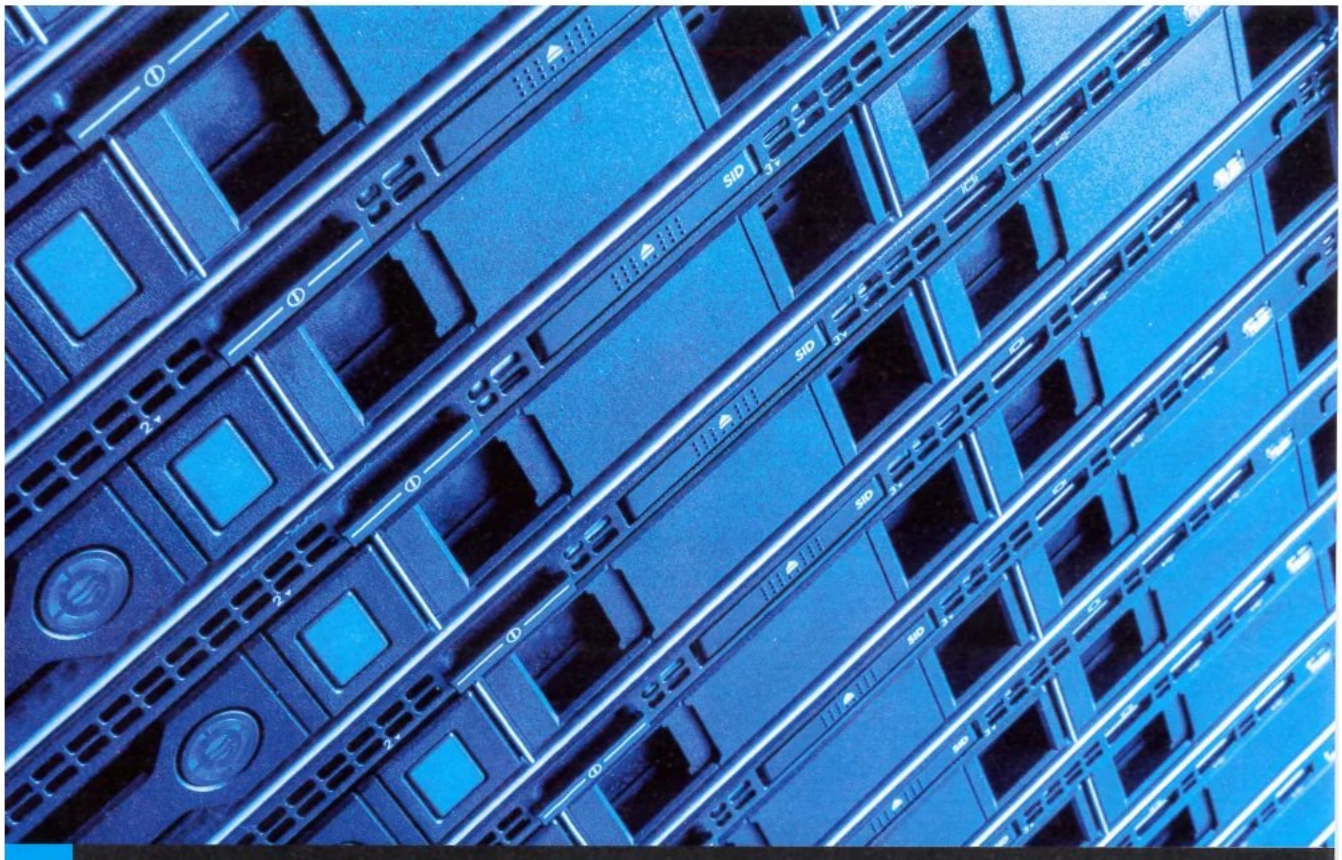
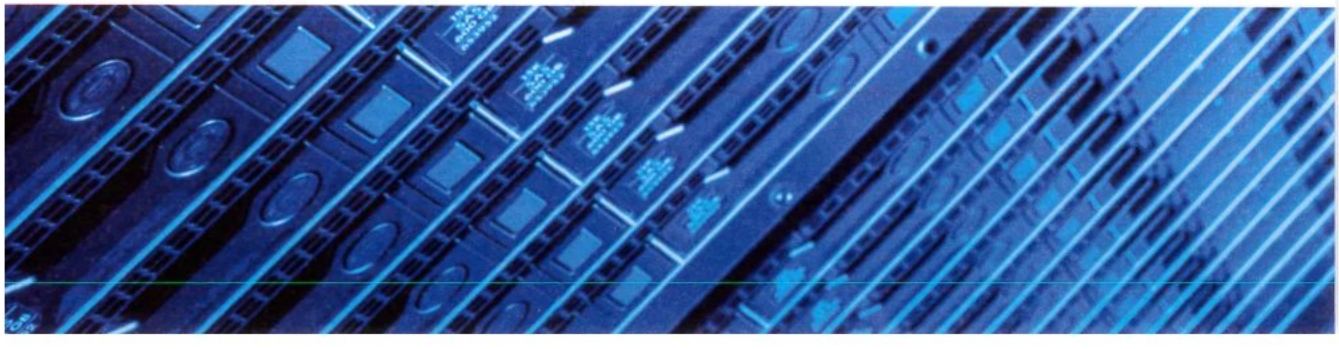


Chapter 10:
Ethics and ownership

Learning objectives

By the end of this chapter you should be able to:

- show understanding of the need for and purpose of ethics as a computing professional
- show understanding of the need to act ethically and the impact of acting ethically or unethically for a given situation
- show understanding of the need for copyright legislation
- show understanding of the different types of software licencing and justify the use of a licence for a given situation
- show understanding of Artificial Intelligence (AI).



10.01 Ethics

You can find a number of definitions of 'ethics'. The following three sentences are examples.

- Ethics is the field of moral science.
- Ethics are the moral principles by which any person is guided.
- Ethics are the rules of conduct recognised in a particular profession or area of human life.

In this book, we will not use the first definition. The third definition is the focus of this chapter. However, the rules of conduct of computer scientists and developers must reflect the moral principles of the second definition. Here are some observations that come to mind when considering moral principles.

Moral principles concern right or wrong. The concept of virtue is often linked to what is considered to be right. What is right and wrong can be considered from one of the following viewpoints: philosophical, religious, legal or pragmatic.

Philosophical debate has been going on for well over 2000 years. Early thinkers frequently quoted in this context are Aristotle and Confucius but there are many more. Religions have sometimes incorporated philosophies already existing or have introduced their own. Laws should reflect what is right and wrong. Pragmatism could be defined as applying common sense.

This chapter is not an appropriate place to discuss religious beliefs, but we should remember that religious beliefs do have to be considered in the working environment. Legal issues clearly impact on working practices but they are rarely the primary focus in rules of conduct. What remains as the foundation for rules of conduct are the philosophical views of right and wrong and the pragmatic views of what is common sense. These will constitute a frame of reference for what follows in this chapter.

10.02 The computing professional

No matter what their particular specialism is, any professional person is expected to act ethically. A professional can receive guidance on ethical behaviour by joining an appropriate professional organisation. Such an organisation will have a code of conduct that will include reference to ethical practice.

For example, the British Computer Society (BCS) has a code of conduct that gives guidance under four headings:

1. Public Interest
2. Professional Competence and Integrity
3. Duty to Relevant Authority
4. Duty to the Profession

The Association for Computing Machinery (ACM) and the Institute of Electrical and Electronics Engineers (IEEE) are both based in the USA but have a global perspective and global influence. The IEEE-CS/ACM Joint Task Force Software Engineering Code of Ethics defines eight principles defined as follows.

1. PUBLIC – Software engineers shall act consistently with the public interest.
2. CLIENT AND EMPLOYER – Software engineers shall act in a manner that is in the best interests of their client and employer consistent with the public interest.
3. PRODUCT – Software engineers shall ensure that their products and related modifications meet the highest professional standards possible.

4 JUDGEMENT – Software engineers shall maintain integrity and independence in their professional judgement.
5 MANAGEMENT – Software engineering managers and leaders shall subscribe to and promote an ethical approach to the management of software development and maintenance.
6 PROFESSION – Software engineers shall advance the integrity and reputation of the profession consistent with the public interest.
7 COLLEAGUES – Software engineers shall be fair to and supportive of their colleagues.
8 SELF – Software engineers shall participate in lifelong learning regarding the practice of their profession and shall promote an ethical approach to the practice of the profession.

Despite the differences in the detail the codes are consistent with regard to the following:

- the public interest or public good is a key concern
- the codes present fundamental principles
- the professional is expected to exercise their own judgement
- the professional should seek advice if unsure.

WORKED EXAMPLE 10.01

Applying ethics to a software engineering scenario

In a real-life scenario there might be many individual clauses that should be considered when a judgement is to be made. For example, let's consider the following scenario.

You are employed by a company that develops software. You are working on a software engineering project to be delivered to a client. One day the project manager states that the project is running behind schedule. As a result, the time allocated for testing of the software will be limited to one week rather than the one month that was stated in the project plan.

As a professional you could be guided in your thinking by referring to the eight principles listed above from the IEEE-CS/ACM Joint Task Force Software Engineering Code of Ethics.

Considering them in turn your thinking might be as follows:

1 You could probably rule out any immediate need to consider public interest.
2 You would recognise that the end result might be the client being delivered a sub-standard product that would reflect badly on the reputation of your employer.
3 You would identify the primary cause of concern as being the likely poor quality product likely to be delivered.
4 You would realise that you needed to make a judgement as to what action, if any, you should take.
5 You might identify the secondary cause of concern as being one of poor management.
6 You would have some concern concerning how delivering a product with many errors would cause your profession to be judged badly but this would not be a primary concern.
7 You would be concerned about your colleagues being put under pressure to deliver in an unrealistically short timescale.
8 You would recognise that this was not an issue relating to your professional development.

(Continued)

You would now consider what action to take. This is where you would need to make a judgement. You might consider four possible scenarios:

1. You decide that this unprofessional behaviour by the project manager must be challenged. Following this challenge the decision is reversed.
2. You decide that this unprofessional behaviour by the project manager must be challenged. However, your protests are ignored.
3. You decide that no challenge is needed because although the testing will not be properly completed there will always be the opportunity to remedy the remaining errors in the code through routine maintenance following product installation.
4. You decide that an immediate protest would be useless but you intend to raise the matter at a later time when the errors in the product have become evident.

The first scenario is the ideal one where the appropriate ethical action leads to a fully-tested product. In the second scenario the professional has acted ethically but this has had no effect. The question is now whether anything else ought to be done. The remaining two scenarios are where unethical behaviour leaves the outcome of an unsatisfactory product being provided for the client.

Discussion Point:

Search the clauses for all eight principles of the IEEE-CS/ACM Joint Task Force Software Engineering Code of Ethics code and identify the ones that mention documentation.

Why is documentation mentioned so many times?

10.03 The public good

In different parts of the IEEE-CS/ACM Joint Task Force Software Engineering Code of Ethics code there is reference to:

- the health, safety and welfare of the public
- the public interest
- the public good
- public concern.

The BCS code has the statement that the professional should:

'have due regard for public health, privacy, security and wellbeing of others and the environment'.

There is no further indication of how these should be interpreted. We can look at some individual cases to illustrate what might be considered.

Fortunately, there are very few examples which have involved loss of life and certainly none where large numbers of deaths were caused. However, there have been a number of incidents where extremely large sums of money were wasted because of rather simplistic errors.

The first example is the Ariane 5 rocket which exploded 40 seconds after blast-off in 1996. Approximately 500 million dollars' worth of investment in development, scientific equipment and launch costs were wasted. The problem was caused by a line of code that tried to convert a 64-bit floating point number into a 16-bit integer. The resulting overflow crashed the program and as a result also the rocket.

The second example also relates to space exploration. The NASA Mars Climate Orbiter project centred on a space probe that was due to orbit Mars to study the climate. The probe reached Mars but unfortunately failed to get into orbit. The cause of the problem was that all

of the software was supposed to use the SI system of units for all calculations. One group of software engineers used the Imperial system of units. This mismatch only caused a problem at the stage when the calculations concerned with achieving orbit around Mars were executed. This time the loss to the public purse was 125 million dollars.

These examples illustrate the public interest in successful software engineering. There is a strong argument that the correct application of the code of ethics with respect to specification, development and testing of software could have saved a lot of money.

A different type of disaster is the system that never gets built. In 2011 the UK government scrapped the National Programme for IT in the NHS (National Health Service), which had been commissioned in 2002. The project failed to produce a workable system. The estimated amount spent on the program was 12 billion pounds. The initial estimated cost was less than three billion pounds. In examples like this the software engineers are not to blame, but if correctly applied, the part of the code of ethics specifically targeted at project management would not have allowed this type of failure to occur.

In the three examples outlined above, the public concern was solely related to the costs associated with a failed project. There was no public concern relating to the ethics of the endeavour itself. In contrast there are many areas associated with computer-based systems where there is public concern about the nature of the endeavour or at least about what it has led to. Here are some examples:

- powerful commercial companies being able to exert pressure on less powerful companies to ensure that the powerful company's products are used when alternatives might be more suitable or less costly
- companies providing systems that do not guarantee security against unauthorised access
- organisations that try to conceal information about a security breach that has occurred in their systems
- private data transmitted by individuals to other individuals being stored and made available to security services
- social media sites allowing abusive or illegal content to be transmitted
- search engines providing search results with no concern about the quality of the content.

There is by no means a consistent public attitude to concerns like this. This makes it difficult for an individual software engineer to make a judgement with respect to public good. Even if the judgement is that a company is not acting in the public good, it will always be difficult for an individual to exert any influence. There are recent examples where individuals have taken action which has resulted in their life being severely affected.

Discussion Point:

This section has deliberately been presented in generalisations. You should carry out a search for some individual examples and then consider actions that could have been taken and justified as being for the public good.

10.04 Ownership and copyright

Copyright is a formal recognition of ownership. If an individual creates and publishes some work that has an element of originality, the individual becomes the owner and can therefore claim copyright. An exception is if the individual is working for an organisation. An organisation can claim copyright for a published work if it is created by one or more individuals that work for the organisation. Copyright cannot apply to an idea and it cannot be claimed on any part of a published work that was previously published by a different individual or organisation.